



Synergistic effects of fish scale–derived nanohydroxyapatite and arginine on enamel surface recovery: a multi-modal in vitro analysis

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Abstract

Dental caries arises from an imbalance between demineralization and remineralization of dental hard tissues, leading to progressive mineral loss and early lesion formation. Fluoride enhances remineralization but mainly affects the enamel surface. Nanohydroxyapatite (n-HA), structurally similar to enamel apatite, may promote mineral deposition, while arginine increases local pH and supports remineralization. This in vitro study evaluated the combined remineralization potential of sustainably sourced, fish scale–derived n-HA and arginine, used alongside fluoride toothpaste, for early enamel lesions using a multimodal analytical approach. Forty human premolars were demineralized for 96 h to produce artificial lesions and randomly allocated into five groups ($n=8$): (1) fluoride toothpaste, (2) 10% n-HA, (3) 10% n-HA+10% arginine, (4) fluoride toothpaste+10% n-HA+10% arginine, and (5) demineralized control. Specimens underwent pH cycling with respective remineralizing slurries for 28 days. Surface changes were evaluated using surface microhardness, scanning electron microscopy (SEM), atomic force microscopy (AFM), and Raman microscopy. The fluoride+n-HA+arginine group showed the greatest increase in microhardness (Vickers Hardness Number [VHN] 367.48 ± 26.95 kgf/mm²), significantly higher than the demineralized control (191.24 ± 13.86 ; $p < 0.001$). The n-HA+arginine group also showed a substantial increase in microhardness (319.10 ± 28.75 kgf/mm²). SEM and AFM analyses revealed greater surface smoothing and pore reduction in the combination group compared with fluoride alone, suggesting enhanced surface recovery and mineral

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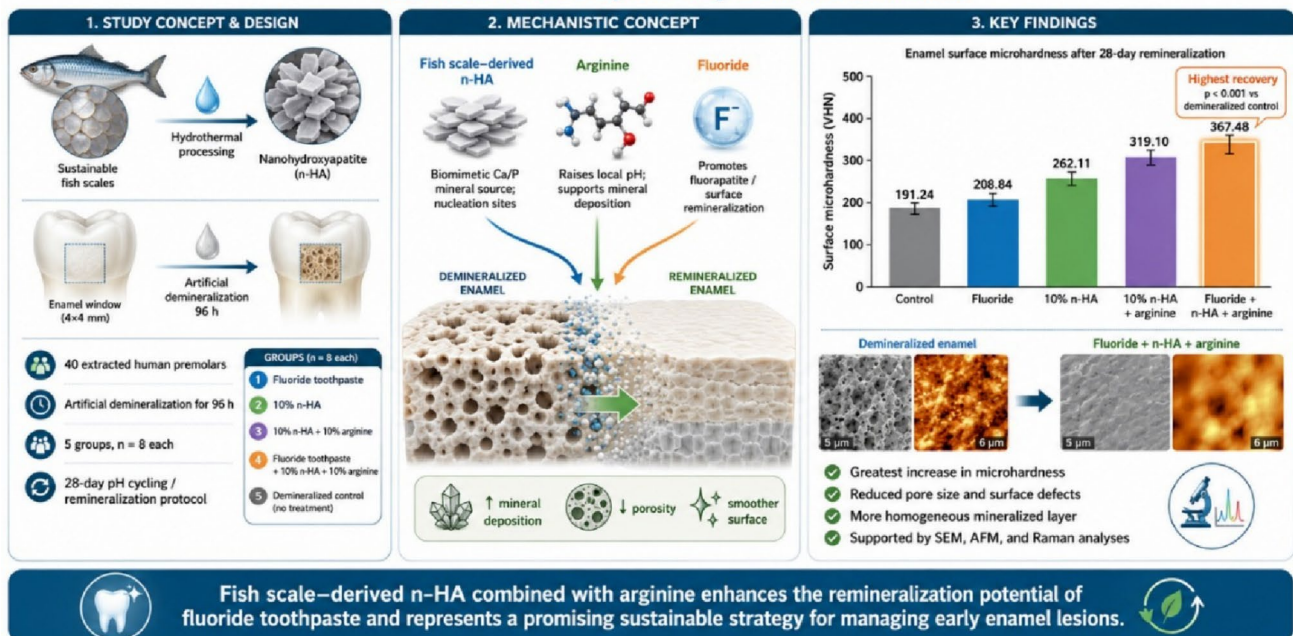
deposition. Fish scale-derived n-HA combined with arginine enhances the in vitro remineralization potential of fluoride toothpaste and may represent a promising strategy for managing early enamel lesions.

Graphical abstract

The combination of fish scale-derived n-HA with arginine increases the enamel surface remineralization capacity of fluoride toothpaste in an in vitro caries model. Forty extracted human premolar teeth were submitted to artificial demineralization and divided into five groups ($n=8$): fluoride toothpaste, 10% n-HA, 10% n-HA + 10% arginine, fluoride toothpaste + 10% n-HA + 10% arginine, and demineralized control. After 28 days of pH cycling, the highest enamel microhardness was achieved by the group treated with fluoride toothpaste + 10% n-HA + 10% arginine (367.48 ± 26.95 VHN) compared with the demineralized control (191.24 ± 13.86 VHN; $p < 0.001$). The results indicated reduced surface porosity and structural imperfections of the enamel surface along with a homogeneous enamel surface, confirmed by morphological and topographical analysis by SEM and AFM. Abbreviations: n-HA, nanohydroxyapatite; VHN, Vickers hardness number; SEM, scanning electron microscopy; AFM, atomic force microscopy

Fish scale-derived nanohydroxyapatite + arginine enhance enamel surface recovery

Multi-modal in vitro analysis of early enamel remineralization



Keywords Tooth remineralization · Dental enamel · Nanohydroxyapatite · Arginine · Dental caries · Fluorides · Fish scale-derived hydroxyapatite · Sustainable biomaterials · Atomic force microscopy

Introduction

Dental caries is a significant global health challenge, causing discomfort, pain, and functional impairment [1, 2]. Thereby, it represents a major threat to dental health, characterized by enamel demineralization and the development of carious lesions [3, 4]. A dynamic balance between demineralization and remineralization on the enamel surface is essential for maintaining the integrity of dental hard tissues [5, 6]. Carious lesions form when the equilibrium shifts towards demineralization [7]. Cariology research primarily aims to maintain optimal concentrations of calcium and

phosphate ions in the oral environment to prevent demineralization and promote remineralization of early carious lesions. The efficacy of fluoride in preventing acid-induced demineralization and facilitating enamel remineralization is well established [8]. Fluoride promotes the formation of fluorapatite crystals within the enamel subsurface, thereby enhancing resistance to acid attack [9]. However, fluoride may form a relatively mineral-rich surface layer that can restrict the penetration of calcium and phosphate ions, which may limit the repair of deeper enamel lesions. Consequently, remineralization is often predominantly confined to the surface, reducing the overall effectiveness in restoring

the entire enamel structure [10]. This limitation underscores the necessity for alternative remineralizing agents that can penetrate deeper enamel layers to deliver essential minerals [11]. Accordingly, this study focuses on developing materials capable of delivering ions into the deeper enamel subsurface, thereby creating a supersaturated oral environment that promotes remineralization.

One such material is nanohydroxyapatite (n-HA), which has morphological and crystallographic similarities to the apatite crystals found in enamel [12]. Due to its nanoscale size, n-HA offers a larger reactive surface area, superior bioactivity, improved solubility, and better biocompatibility than conventional hydroxyapatite [13]. Importantly, fish scale-derived n-HA differs from synthetically produced n-HA in that it is a biogenic, carbonate-substituted apatite carrying trace ions that resemble the mineral phase of biological hard tissues, which may favor its integration with enamel while offering a sustainable, low-cost source of nanohydroxyapatite. In this study, n-HA derived from fish scales was used, as its biological origin may facilitate natural incorporation into and integration with the remineralization process [14]. Its bio-acceptability enhances its ability to bond with natural tooth structure, making it a promising candidate for improving remineralization and dental health outcomes. In addition, using fish scale-derived n-HA supports sustainability by valorizing biological waste, aligns with environmentally sustainable practices, and provides a cost-effective, readily available resource for dental applications. This strategy aligns with the growing emphasis on environmental sustainability in dentistry and the increasing use of naturally derived, biowaste-sourced hydroxyapatite for dental and biomedical applications [15–18]. Furthermore, arginine, a naturally occurring amino acid found in dietary proteins, has been identified as a potent pro-remineralizing agent [19]. It is metabolized by arginolytic oral bacteria, producing ammonia, thereby increasing oral pH, which supports mineral deposition [20]. Research suggests that supplementing fluoride toothpaste with arginine enhances its biofilm-modifying properties and improves overall dental health [21]. Arginine's ability to maintain an alkaline environment helps counteract demineralization, thereby supporting improved morphology [22].

In the present study, we evaluated the combined effect of fish scale-derived n-HA, arginine, and fluoride toothpaste on enamel surface recovery. We hypothesized that the combined application of n-HA and arginine with fluoride would result in greater increases in enamel surface microhardness and improved surface morphology compared with fluoride alone. This approach aims to overcome the limitations of surface-dominated fluoride action by enhancing ion availability and promoting more effective subsurface mineral deposition.

Materials and methods

Sample selection and preparation

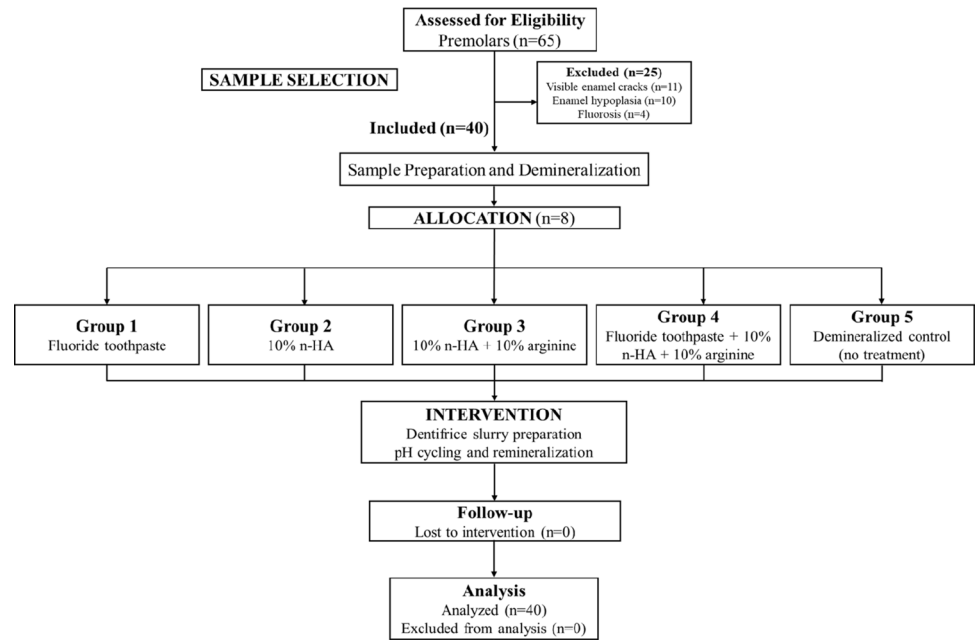
Forty sound permanent human premolars, extracted for orthodontic purposes and exhibiting no visible enamel defects, were collected and prepared for this study. Ethical approval for this study was obtained from the Institutional Review Board of Pushpagiri Institute of Medical Sciences and Research Centre, Thiruvalla, Kerala, India (Approval No.: IRB/02/14/2024).

The natural n-HA used in this study was derived from fish scales through a hydrothermal process. In brief, cleaned fish scales were subjected to a controlled hydrothermal treatment in an aqueous medium at elevated temperature and pressure, converting the native biogenic calcium phosphate into crystalline nanohydroxyapatite while preserving the carbonate and trace-ion substitutions characteristic of biological apatite; the physicochemical and crystallographic properties of this fish scale-derived n-HA have been characterized in detail previously [14]. The apatitic nature and surface morphology of this fish scale-derived hydroxyapatite were documented in the source characterization studies using techniques including atomic force microscopy; a dedicated in-study physicochemical re-characterization (particle-size distribution, X-ray diffraction, and Fourier-transform infrared spectroscopy) was not undertaken and is acknowledged as a study limitation. This biological origin was selected to leverage its crystallographic similarity to human enamel and its inherent biocompatibility, thereby facilitating seamless integration into the remineralization process. Following thorough cleaning and disinfection with 5% sodium hypochlorite, the teeth were preserved in artificial saliva to maintain their structural integrity. The root portions were removed at the cemento-enamel junction using a diamond disk with continuous water cooling. Crown sections were mounted in acrylic resin molds measuring 25.4 mm in diameter and 15 mm in height, ensuring that the buccal surface was parallel to the floor. A 4 × 4 mm working window was delineated on the buccal surfaces, and the adjacent areas were coated with acid-resistant nail polish (Revlon, USA) to confine the demineralization and remineralization processes to the specified region [23]. The study flowchart is depicted in Fig. 1.

Demineralization cycle

The demineralization solution was prepared using 50 mM acetic acid, 2.2 mM calcium nitrate, 2.2 mM potassium dihydrogen phosphate, and 0.1 ppm sodium fluoride, with the pH adjusted to 4.5. Each tooth was continuously immersed in this solution for 96 h to create artificial carious lesions

Fig. 1 Flowchart of the experimental procedure and sample allocation



with an approximate depth of 100–120 μm [24]. Following demineralization, the samples were allocated into five treatment groups: Group 1 received fluoride toothpaste; Group 2 received 10% n-HA; Group 3 received 10% n-HA combined with 10% arginine; Group 4 received fluoride toothpaste combined with 10% n-HA and 10% arginine; and Group 5 served as the demineralized control and received no treatment. Specimens were randomly assigned to the five groups using a computer-generated randomization sequence, and the operator who performed the microhardness measurements and surface imaging was blinded to group allocation. While the cited protocol by *Erdwey et al.* used a 7-day demineralization period, the present study employed a 4-day (96-h) immersion period. A pilot study conducted prior to the main experiment confirmed that this 96-h exposure consistently produced artificial carious lesions with a depth of approximately 100–120 μm , which was sufficient to evaluate the subsurface remineralization potential of the tested formulations. The 96-h (4-day) demineralization protocol was selected because preliminary testing demonstrated that it reliably produced early, subsurface lesions of the target depth (approximately 100–120 μm) suitable for evaluating remineralization, while limiting progression toward deeper, cavitated lesions that respond less predictably to remineralizing agents. The depth and reproducibility of the lesions were verified on the pilot specimens before the main experiment was commenced.

Dentifrice slurry formulation

Dentifrice slurries for the five experimental groups were prepared by mixing each formulation with distilled water at

a 1:3 (w/w) ratio. The dentifrice slurries were centrifuged at 4,000 rpm for 15 min to collect the supernatant for use in the pH cycling protocol. This method was intentionally selected to isolate the soluble ionic fraction, specifically the dissociated calcium, phosphate, and fluoride ions, to evaluate their direct impact on mineral deposition. While centrifugation does not occur in the oral cavity, this standardized laboratory approach allows for a controlled assessment of the chemically active components available for remineralization, free from the mechanical interference of larger insoluble particles. For Group 1, fluoride toothpaste was mixed with distilled water in the specified ratio until homogeneous. Group 2 was prepared by adding fish scale–derived n-HA to distilled water and stirring until fully dispersed. Group 3 was prepared by adding 10% n-HA and 10% arginine to distilled water and mixing thoroughly. Group 4 was prepared by combining fluoride toothpaste, n-HA, and arginine with distilled water, then stirring until a homogeneous slurry was formed. Group 5 was left untreated to serve as the control group (Fig. 1). All mixtures were vortexed for 60 s at room temperature. Each slurry was subsequently centrifuged at 4,000 rpm for 15 min at 25 °C. Following centrifugation, the supernatant was collected for use in the pH-cycling experiments. The concentrations of 10% n-HA and 10% arginine were selected to ensure a supersaturated environment with respect to calcium and phosphate ions. This concentration was intended to maximize the slurry's remineralization potential by providing a sufficient reservoir of ions to counteract demineralization during the 28-day pH-cycling protocol. The 10% concentration was further informed by prior dose–response work showing that nanohydroxyapatite enhances enamel remineralization in

a concentration-dependent manner, with higher concentrations yielding greater mineral gain [35]; accordingly, 10% was selected to ensure ionic supersaturation while maintaining slurry homogeneity, whereas lower concentrations would be expected to produce a smaller remineralizing effect. The centrifugation step was standardized in-house for the present study to isolate the soluble ionic supernatant; as no identical published protocol was available, the speed and duration (4,000 rpm for 15 min) were optimized during the pilot phase to obtain a clear, particle-free supernatant.

The fluoride toothpaste used in this study was Sensodyne Multi Care+Whitening (manufactured by GlaxoSmithKline, Brentford, United Kingdom), containing 1450 ppm fluoride as sodium fluoride (0.315% w/w). The active ingredients included potassium nitrate (5% w/w), sorbitol, hydrated silica, glycerin, PEG-8, cocamidopropyl betaine, flavoring agents, xanthan gum, titanium dioxide, and sodium saccharin. This standard formulation was selected

to represent a conventional over-the-counter dentifrice for comparison with the experimental n-HA and arginine groups. It is acknowledged that this whitening variant may contain additional abrasive and whitening components, as well as a sodium fluoride concentration toward the higher end of the over-the-counter range; the potential influence of these factors on the surface response is considered in the study's limitations.

Remineralization

Each sample was treated with its designated remineralizing agent slurry for 2 min twice daily using a soft toothbrush as part of the pH-cycling protocol. The regimen consisted of two immersion cycles per day: 3 h in a demineralizing solution followed by 3 h in artificial saliva. This alternating demineralization–remineralization regimen was designed to approximate the dynamic pH fluctuations that occur in the oral environment following dietary acid challenges; pH-cycling models of this type are well-established in vitro surrogates for the caries process and have been shown to reproduce the dose- and pH-response behavior observed in clinical studies [25]. Following each cycle, samples were brushed with their assigned remineralizing agent and stored overnight in artificial saliva (pH 7.4) at 37 °C. This procedure was repeated daily for 28 days, with the artificial saliva replaced every 24 h with 20 mL of freshly prepared, pH-regulated solution [26]. To monitor potential surface loss or abrasion caused by the soft-bristle toothbrushing protocol, samples were analyzed using atomic force microscopy (AFM) to evaluate changes in surface topography and roughness. An overview of the study design is provided in Fig. 2.

Surface microhardness measurement

Surface microhardness testing was conducted at Saveetha Dental College and Hospital, Chennai, India, utilizing a Vickers microhardness tester (Shimadzu HMV-G31DT, Japan). Prior to the pH-cycling phase, all 40 specimens underwent baseline microhardness testing after the 96-h demineralization cycle, ensuring standardized initial lesion conditions before group allocation. Indentation sites were selected using a 10× objective, and diamond-shaped indentations were created by applying a 100 g load for 30 s. The indentations were observed and measured on a display monitor at 40× magnification. Post-treatment enamel microhardness for each sample group was evaluated using the same instrument, and the results were recorded as Vickers hardness numbers (VHN) [27, 28]. Baseline microhardness was measured after demineralization to standardize initial lesion

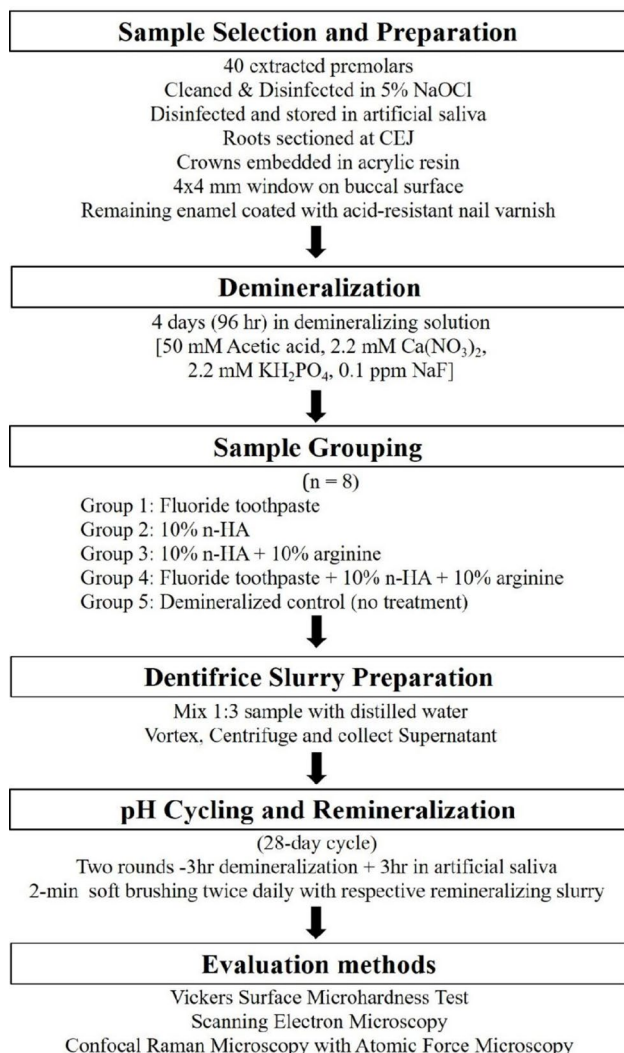


Fig. 2 Schematic representation of the experimental protocol and evaluation methods

conditions prior to group allocation; therefore, comparisons were based on post-treatment VHN values.

Scanning electron microscopy (SEM) analysis

SEM was employed to examine the enamel surface morphology and identify regions of porosity within the samples. For sample preparation, the SEM holder was coated with double-sided carbon tape, with one side attached to the holder and the other to the sample. The assembled samples and holder were placed in a sputter coater and subjected to vacuum for 30 min. A plasma coating of gold (Au) was applied for 3 s, following the procedure outlined by Mulyawan et al. [29]. The SEM stage and specimens were subjected to an additional 5-min vacuum treatment before imaging. Images were acquired at magnifications of 500×, 2,000×, and 5,000× to evaluate surface porosity and morphological changes. Pore dimensions were assessed by drawing measurement lines across selected pores using the SEM software, and the surface characteristics were analyzed qualitatively.

Confocal Raman microscopy with AFM

Confocal Raman imaging integrates Raman spectroscopy with confocal microscopy, yielding high-resolution spectral data at each pixel and achieving lateral resolution at the diffraction limit, defined as approximately half the excitation wavelength [30]. This configuration enables precise depth resolution, facilitating the acquisition of three-dimensional (3D) Raman images and depth profiles. Correlative Raman imaging and AFM were conducted using the WITec alpha300RA system (WITec GmbH, Ulm, Germany) under ambient conditions (24 ± 2 °C). AFM was employed to characterize sample topography with nanometer-scale resolution. Beyond high-resolution topographic imaging, AFM was also utilized to evaluate local surface properties, including adhesion and stiffness. Samples were analyzed in digital pulsed force mode (DPFM) to distinguish regions with varying mechanical properties.

Results

Enamel surface microhardness analysis

The VHN was used to evaluate enamel surface microhardness, with significant variations observed among the sample groups due to compositional and structural differences between the tested materials [31]. Group 4 (fluoride toothpaste + n-HA + arginine) showed the highest mean VHN value (367.48 ± 26.95 kgf/mm²). Group 3 (n-HA + arginine)

also showed a substantial increase in microhardness, with a mean value of 319.10 ± 28.75 kgf/mm², suggesting a beneficial effect of arginine in combination with n-HA. Group 2 (n-HA alone) exhibited a mean value of 262.11 ± 22.56 kgf/mm², suggesting that n-HA contributes to increased surface microhardness, which appears to be enhanced when combined with additional active components. Group 1 (fluoride toothpaste alone) showed a lower mean value of 208.84 ± 13.29 kgf/mm², suggesting comparatively limited improvement in surface microhardness relative to the n-HA-containing groups. The demineralized control group (Group 5) had the lowest mean value (191.24 ± 13.86 kgf/mm²), representing the baseline hardness of demineralized enamel. Overall, these findings indicate that the combination of n-HA and arginine results in greater increases in enamel surface microhardness compared with fluoride toothpaste alone, which may be associated with improved surface mineralization.

Enamel surface morphology assessed by SEM

SEM examination of the enamel surfaces (Figs. 3a–e) provided qualitative insights into the effects of the different remineralizing treatments. The SEM image of Group 5 (Fig. 3a), the demineralized control group, showed a rough and irregular enamel surface with numerous pores and microstructural defects, indicating significant mineral loss and susceptibility to further demineralization.

In Group 1 (Fig. 3b), macropores and surface irregularities were still evident, consistent with limited surface mineral deposition. In contrast, Group 2 (Fig. 3c) exhibited a noticeable reduction in pore size and number, suggesting partial surface mineral deposition and improved enamel surface integrity.

Group 3 (Fig. 3d) suggested further improvements in enamel morphology, with more pronounced surface smoothing and mineral deposition, consistent with enhanced surface mineral deposition compared with the previous groups. The most pronounced effects were observed in Group 4 (Fig. 3e), which used a combination of fluoride toothpaste, n-HA, and arginine. SEM images of this group showed a marked reduction in pore size and density, suggesting substantial surface mineral deposition and a smoother, more homogeneous mineralized layer, with pronounced surface smoothing and recovery.

AFM analysis of enamel surface topography

Figures 4a–e show the AFM images of all groups, including the demineralized teeth. The AFM images of Group 5 (Fig. 4a) served as a baseline, illustrating the initial state of demineralization with a rough surface and pronounced

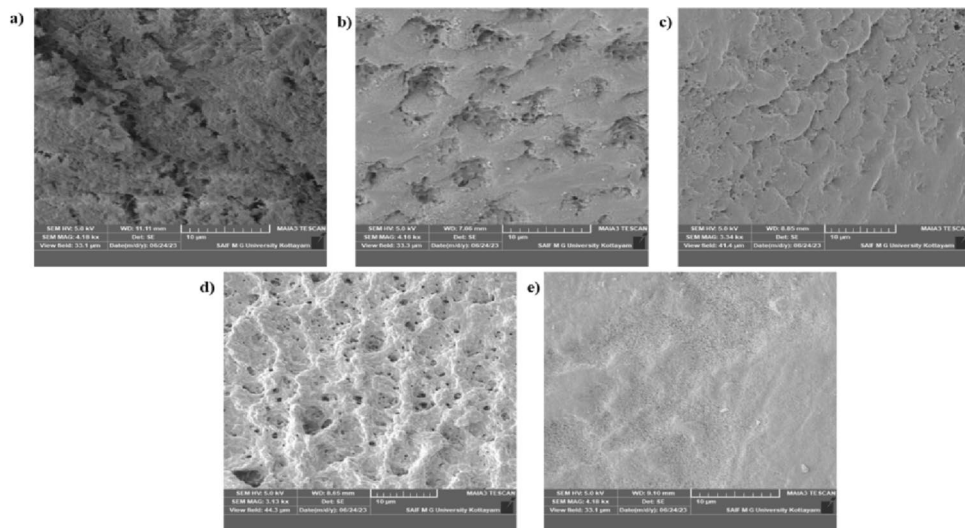
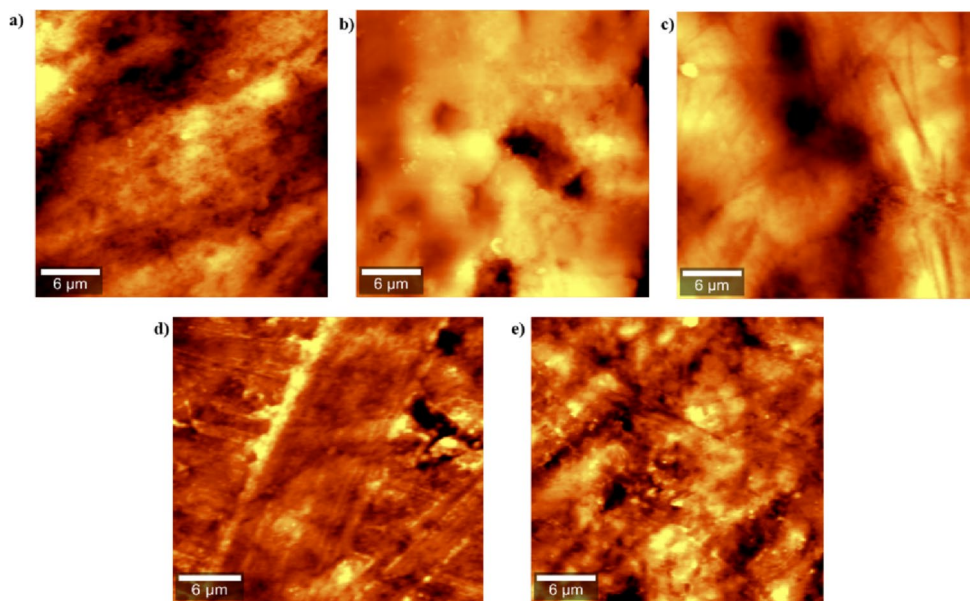


Fig. 3 Representative SEM micrographs of enamel surfaces after remineralization treatments ($\times 5,000$; scale bar = 10 μm). **a** Demineralized control (Group 5) showing a highly porous and irregular enamel surface. **b** Fluoride toothpaste (Group 1) showing persistent surface porosity. **c** 10% n-HA (Group 2) suggesting partial surface mineral

deposition with reduced pore size. **d** 10% n-HA + 10% arginine (Group 3) showing improved surface smoothing and mineral deposition. **e** Fluoride toothpaste + 10% n-HA + 10% arginine (Group 4) exhibiting a dense, homogeneous surface with minimal porosity, suggesting the greatest surface mineral deposition effect

Fig. 4 Representative AFM images of enamel surface topography following different remineralization treatments (scale bar = 6 μm). **a** Demineralized control (Group 5) showing a rough surface with pronounced irregularities and mineral loss. **b** Fluoride toothpaste (Group 1) showing limited surface repair with persistent surface defects. **c** 10% n-HA (Group 2) suggesting partial surface mineral deposition with reduced surface roughness. **d** 10% n-HA + 10% arginine (Group 3) showing improved surface smoothing and mineral deposition. **e** Fluoride toothpaste + 10% n-HA + 10% arginine (Group 4) exhibiting the smoothest and most homogeneous enamel surface, suggesting the greatest surface mineral deposition effect



irregularities, indicative of enamel damage and mineral loss. In Group 1 (Fig. 4b), limited mineral deposition was observed, suggesting a modest surface mineral deposition effect of fluoride toothpaste alone on demineralized enamel [32]. In Group 2 (Fig. 4c), AFM images showed greater surface improvement compared with Group 1. The addition of 10% n-HA appeared to positively affect the enamel surface, with a smoother texture and reduced surface roughness, suggesting enhanced mineral deposition. Group 3 (Fig. 4d), which included 10% n-HA and 10% arginine, suggested further improvements in surface morphology, suggesting that the combination of these agents enhanced surface mineral

deposition through a synergistic effect. Finally, the most pronounced surface mineral deposition was observed in Group 4 (Fig. 4e), where fluoride toothpaste was combined with n-HA and arginine. The synergistic action of fluoride, n-HA, and arginine resulted in a markedly smoother enamel surface with minimal surface irregularities, suggesting enhanced surface smoothing and reduced roughness [33].

Table 1 Surface microhardness of different treatment groups, with $n=8$ samples per group, compared by one-way ANOVA

Group	mean VHN (kgf/mm ²)	SD	95% CI (kgf/ mm ²)	<i>p</i> -value
1: Fluoride toothpaste	208.84	13.29	197.73– 219.95	<0.001
2: n-HA 10%	262.11	22.56	243.25– 280.97	
3: n-HA 10% + arginine	319.10	28.75	295.06– 343.14	
4: Fluoride tooth- paste + n-HA 10% + arginine	367.48	26.95	344.95– 390.01	
5: demineralized control	191.24	13.86	179.65– 202.83	

VHN Vickers hardness numbers, SD standard deviation, 95% CI 95% confidence interval for the mean

Statistical analysis of surface microhardness differences

A one-way ANOVA was performed to compare surface microhardness among the experimental groups (Table 1). The Shapiro–Wilk test indicated that the assumption of normality was met for each group. Levene’s test for homogeneity of variances showed that the assumption of equal variances was not violated ($F(4, 35)=2.99$, $p=0.07$). The ANOVA results were statistically significant ($F(4, 35)=90.12$, $p<0.001$), indicating significant differences in surface microhardness among the experimental groups. Post-hoc analysis using Tukey’s HSD test showed that Group 4 (fluoride toothpaste + n-HA + arginine) with a mean of 367.48 (SD=26.95) showed significantly higher surface microhardness compared with the demineralized control group (Group 5) (mean=191.24, SD=13.86; $p<0.001$). This finding highlights the enhanced efficacy of surface mineral deposition by the combined formulation containing fluoride, n-HA, and arginine. The corresponding 95% confidence intervals (CIs) for the group means are reported in Table 1; notably, the 95% CI for the combined fluoride + n-HA + arginine group (Group 4: 344.95–390.01 kgf/mm²) did not overlap with that of the demineralized control (Group 5: 179.65–202.83 kgf/mm²), reinforcing the statistical significance of this difference. The overall effect size was large ($\eta^2 = 0.91$, derived from the ANOVA), indicating that group allocation accounted for approximately 91% of the variance in surface microhardness.

Discussion

The present study evaluated the surface mineral deposition potential of a combination of naturally derived n-HA from fish scales and arginine in conjunction with fluoride-containing

toothpaste by assessing changes in enamel microhardness and surface characteristics using AFM and SEM. The n-HA used in this study was derived from fish scales through a hydrothermal process, as characterized by *Binsi et al.* [14]. Group 4, which combined fluoride toothpaste, n-HA, and arginine, showed the highest mean VHN value, indicating a synergistic effect that enhances the protective and reparative properties of these components. *Poggio et al.* confirmed that adding n-HA to fluoride toothpaste significantly improved enamel surface microhardness and was associated with enhanced mineral deposition compared with fluoride toothpaste alone [34].

The inclusion of arginine further enhanced surface mineral deposition and associated microhardness changes by stabilizing calcium ions, promoting mineral deposition, and modulating the activity of cariogenic bacteria. Arginine’s ability to generate an alkaline microenvironment and facilitate calcium and phosphate deposition on enamel, as shown by *Huang et al.* and *Uraz et al.* [35, 36] plays a critical role in optimizing the remineralizing potential of n-HA. *Yesilyurt et al.* reported increased surface microhardness following the use of arginine-containing toothpaste [37], and *Ebadifar et al.* found that n-HA toothpaste exhibited superior remineralizing effects compared with fluoride toothpaste [38]. *Geeta et al.* also suggested that n-HA crystals significantly improved enamel surface microhardness following the remineralization of initial carious lesions [39].

These findings support previous reports on the susceptibility of demineralized enamel [40, 41], suggesting that enamel subjected to surface mineral deposition processes becomes more resistant to subsequent mineral loss [42]. The significant recovery of surface microhardness in Group 4 reflects substantial mineral recovery approaching the hardness of sound enamel. While this level of recovery may appear notable for an in vitro model, it is consistent with the synergistic interaction of high-concentration (10%) n-HA, arginine, and fluoride, which collectively may facilitate enhanced mineral deposition beyond the immediate surface.

Furthermore, the multi-modal surface analyses using AFM and SEM confirmed that the soft-bristle toothbrushing protocol did not induce significant lesion abrasion or surface loss. Instead, AFM imaging revealed a smoother enamel surface with minimal irregularities, suggesting that the rate of surface mineral deposition exceeded any potential mechanical wear associated with the daily brushing regimen. The surface-sensitive analyses (SEM, AFM, and correlative confocal Raman–AFM imaging) consistently demonstrated progressive pore reduction and surface smoothing across the n-HA- and arginine-containing groups, with the most homogeneous, densely mineralized surface observed in Group 4. These qualitative observations are consistent with previous atomic force microscopy and

electron microscopy studies of enamel remineralization, in which casein phosphopeptide–amorphous calcium phosphate and nanohydroxyapatite agents produced comparable reductions in surface roughness and pore size [33, 34, 39]. The convergence of independent surface-analytical techniques strengthens the interpretation that the combined formulation promotes greater surface mineral deposition than fluoride alone, and underscores the value of multi-modal characterization of naturally derived hydroxyapatite materials [15, 16].

This study contributes to the field of preventive dentistry by suggesting the synergistic benefits of combining fish scale–derived n-HA and arginine with fluoride toothpaste. The smooth, pore-reduced enamel surface observed in Group 4 suggests an effective surface mineral deposition process, consistent with the findings of *Acevedo et al.* regarding the combined benefits of multiple remineralizing agents [43]. The hydrothermal process used to derive n-HA from fish scales enhances its biocompatibility and may facilitate the release of trace minerals and ions, which can integrate into the demineralized enamel surface.

A critical consideration in formulating the tested slurries is the potential chemical interaction between nanohydroxyapatite and fluoride in aqueous environments. In this study, however, the synergistic effect observed in Group 4 suggests that the slurry preparation and centrifugation method used to isolate the soluble ionic supernatant provided sufficient ionic availability to initiate surface mineral deposition. Future commercial development would likely require stabilized or anhydrous formulations to maintain long-term stability and clinical applicability.

Although the sample size was sufficient for statistical analysis ($n=8$ per group), it may not fully represent the variability present in clinical populations or differing oral conditions, thereby limiting the generalizability of the findings. Since a formal a priori sample size calculation was not performed, this study should be considered exploratory. Future research should include longer observation periods and larger, more diverse sample populations to validate these results. Additionally, investigating different concentrations of n-HA and arginine in combination with fluoride toothpaste could provide valuable insights into optimizing surface mineral deposition efficacy for early enamel carious lesions. Since there is no universally accepted gold standard for in vitro enamel remineralization models, the demineralized control group in this study is used as an in vitro reference rather than as a direct clinical gold standard. Consequently, while the multi-modal surface analyses (SEM and AFM) corroborate the microhardness findings, the results should be considered a proof-of-concept that requires validation in larger and more clinically representative studies. It should be noted that surface microhardness is an indirect measure,

and while it reflects changes in mineral content, it does not directly confirm subsurface remineralization. Several additional limitations should be acknowledged. First, the present study relied on surface microhardness together with topographical and morphological imaging (SEM, AFM, and correlative confocal Raman–AFM); complementary physicochemical characterization such as Fourier-transform infrared (FTIR) spectroscopy, full Raman spectral analysis, and X-ray diffraction (XRD), together with biological and microbiological assays, was not performed. These analyses, which have been used to characterize naturally derived hydroxyapatite in related work [15, 16], would further confirm the composition, crystallinity, and biological behavior of the remineralized layer and represent an important direction for future work. Second, the fluoride dentifrice used was a whitening formulation that may contain additional abrasives and a sodium-fluoride concentration toward the higher end of the over-the-counter range, which could have influenced the surface response. Third, although the sample size ($n=8$ per group) was adequate to detect statistically significant differences, it was not derived from a formal a priori power calculation; future studies should incorporate power-based sample-size estimation. Fourth, the surface morphology and topography were analyzed descriptively; quantitative parameters such as surface roughness (R_a , R_q) and pore-size distribution were not extracted from the SEM and AFM data and should be incorporated in future analyses to enhance analytical rigor. Fifth, surface microhardness reflects predominantly surface-level changes, and complementary subsurface techniques, such as cross-sectional microhardness profiling or transverse microradiography, would be required to confirm subsurface mineral recovery. Finally, because the experimental design was not a full factorial (arginine-alone and fluoride-plus-n-HA groups were not included), a formal statistical interaction analysis could not be performed; the description of the combined effect as synergistic is therefore used in a descriptive sense, and the interaction should be confirmed in a dedicated future factorial study.

The experimental design includes pH cycling and Vickers microhardness testing. These methods align with established in vitro methodologies typically used to evaluate enamel surface alterations. This supports the methodological validity of the study. However, these models do not fully replicate the complexity of the oral environment, including salivary dynamics, biofilm interactions, and patient-specific variability. From a translational perspective, n-HA and arginine are both widely reported to be biocompatible. Both are already incorporated into commercially available oral care products. However, more in vivo and clinical studies are needed to confirm their long-term safety and efficacy.

Conclusion

The study suggests that incorporating naturally derived n-HA from fish scales and arginine into fluoride toothpaste significantly improves enamel surface microhardness and surface characteristics, suggesting enhanced mineral deposition. This combination yields greater improvements in enamel microhardness and surface properties than conventional fluoride treatments. n-HA, which closely resembles natural hydroxyapatite, and arginine act synergistically to promote surface mineral deposition. The findings indicate that incorporating these natural compounds into dental products may enhance caries prevention, reduce associated costs, and improve patient outcomes. From a translational and public health perspective, the valorization of fishery byproducts as a source of n-HA represents a potentially low-cost, scalable, and environmentally sustainable approach to caries prevention that may be especially relevant in resource-limited settings; formal analyses of cost-effectiveness and large-scale manufacturability remain to be conducted.

Author contributions N.G.T., A.M.X., S.A., and A.C. contributed to the conceptualization and study design. N.G.T., V.T.O., A.M.X., R.D., P.K.B., B.M.M., and S.M.S. performed data curation. P.K.B., S.V., S.N., N.M.A., and B.M.M. conducted the formal analysis and interpretation of the data. Resources were provided by P.K.B., R.D., S.V., S.N., N.M.A. and A.C. Validation was carried out by N.G.T., V.T.O., P.K.B., N.M.A., S.A., G.S., and A.C. The original manuscript draft was prepared by N.G.T., A.M.X., B.M.M., and S.M.S. A.C., S.A., and G.S. were responsible for manuscript review, editing, and final approval. All authors have read and approved the final manuscript.

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Data availability The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare no conflicts of interest.

Ethical approval The study received ethical approval from the Institutional Review Board of Pushpagiri Institute of Medical Sciences and Research Centre, Thiruvalla, Kerala, India (Approval No.: IRB/02/14/2024).

Disclosure AI-assisted language editing was used to improve readability; no data extraction, statistical analysis, or interpretation was performed by AI. All content was reviewed and approved by the authors.

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